

HELP DESK MANAGEMENT SYSTEM

Project Submission Report & Code Documentation

ASP.NET Core 8 Web API • MVC • Entity Framework Core

Name	Rajrup Roy Chowdhury
Registration No.	IN26010404
Project	Help Desk Management
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Repository URL	Github

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1. Executive Summary & Project Objective

The objective of this project was to build an internal Help Desk system that lets employees raise support tickets, and lets help desk agents track, assign, update and close them.

The solution is built as two ASP.NET Core applications. The Web API handles all the data and the business rules, and the MVC web application provides the user interface. The MVC application does not connect to the database at all — it calls the Web API over HTTP for everything. This keeps the two layers separate, so the same API could later be used by a mobile app or another client.

Main features delivered:

- Create, view, edit and delete support tickets (full CRUD)
- Search, filter by status / priority / category, sort by column and page through results
- SLA due dates calculated automatically from the ticket priority
- Comments on each ticket, plus an automatic history of every status, priority and assignment change
- A dashboard showing summary counts, category and priority breakdowns and overdue tickets
- Validation on both the client side and the server side

2. System Architecture & Technical Stack



2.1 Technology Used

Layer	Technology
Backend Framework	ASP.NET Core 8.0 Web API
Frontend Framework	ASP.NET Core 8.0 MVC (Razor Views)
Database	SQLite (default); can switch to SQL Server via configuration
ORM	Entity Framework Core 8.0
API Documentation	Swagger / OpenAPI (Swashbuckle)
Unit Testing	xUnit with FluentAssertions
Styling	Custom CSS (no framework)

2.2 Project Structure

Project	Responsibility
HelpDesk.Contracts	Shared enums, DTOs and validation attributes used by both applications
HelpDesk.Api	Controllers -> Services (business rules) -> Repositories (data access)
HelpDesk.Mvc	Controllers -> API client -> Razor views
HelpDesk.Tests	Unit tests for the service and repository layers

The Repository Pattern was used to separate the database code from the business logic, and a Service layer was added on top so that the controllers stay small and only handle HTTP concerns.

3. Application Screenshots & Visual Proof

Paste each screenshot into the box below its heading, then delete the dashed box. Start both applications first (see Section 7).

Figure 1: Web API running with Swagger documentation

Capture: <http://localhost:5285/swagger>

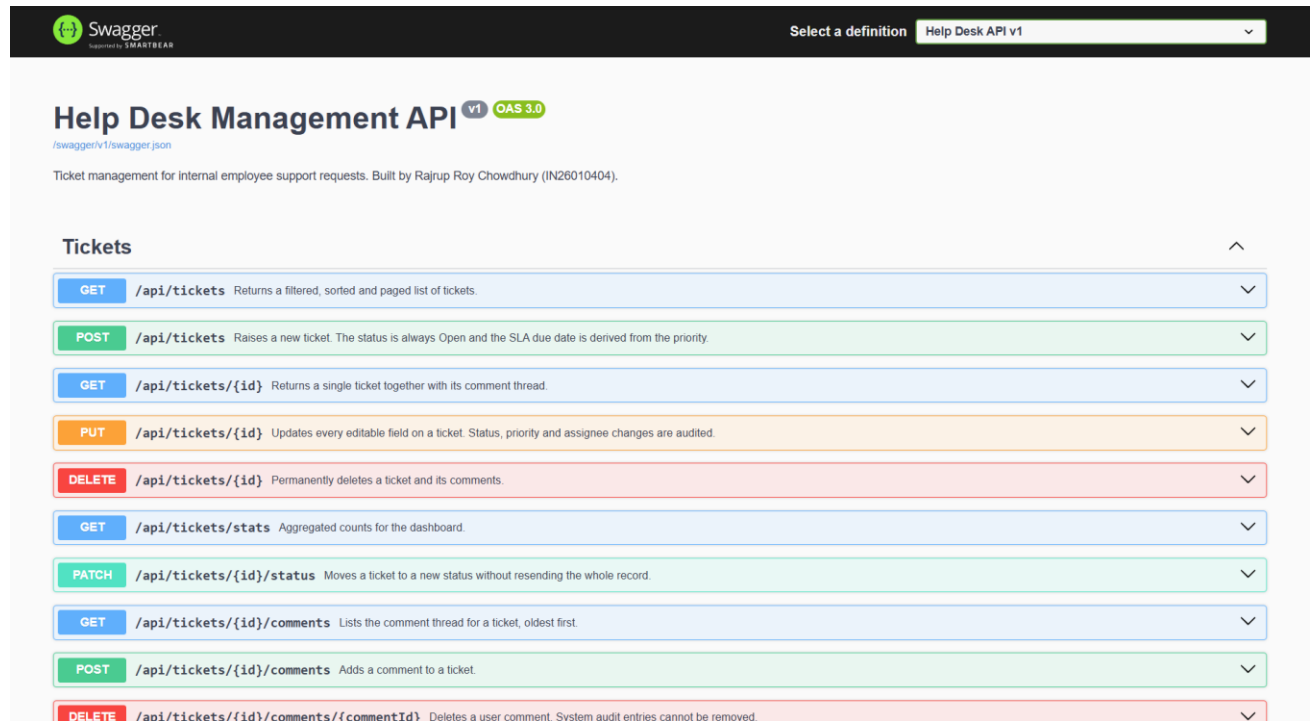


Figure 2: Help Desk Dashboard

Capture: <http://localhost:5185>

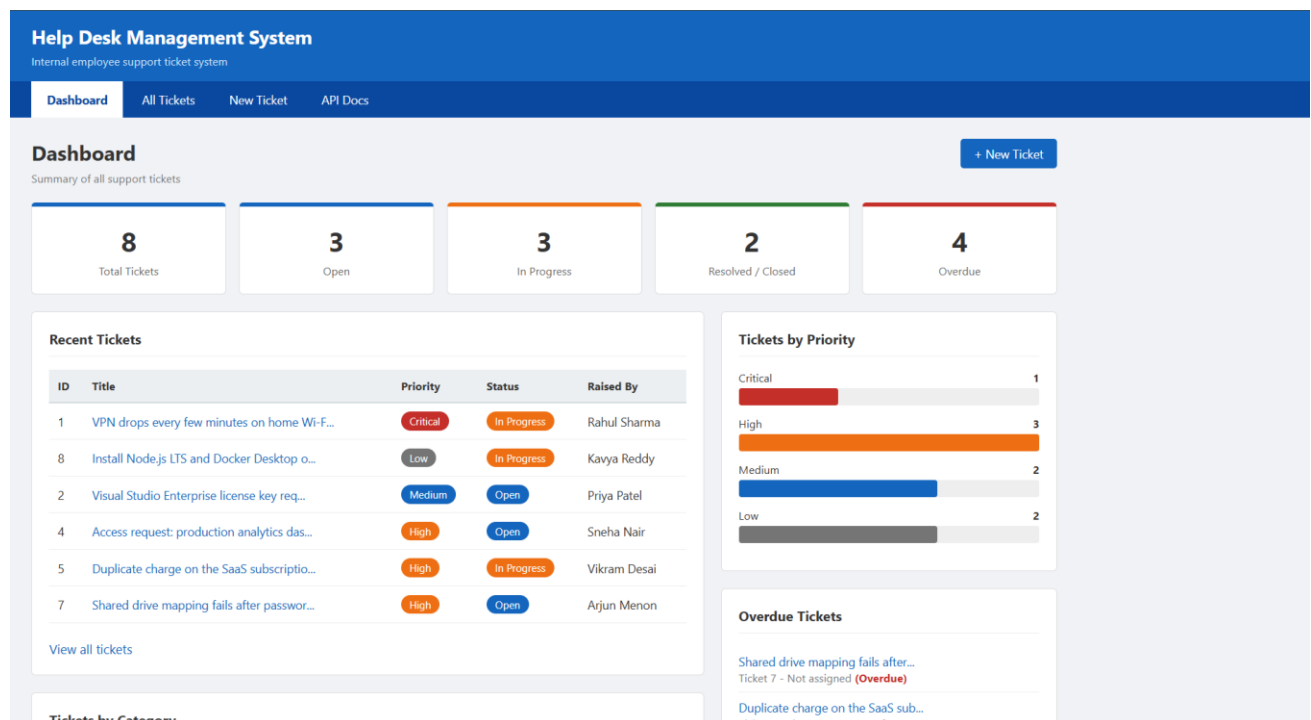


Figure 3: All Tickets list

Capture: <http://localhost:5185/Ticket/Index>

Help Desk Management System

Internal employee support ticket system

[Dashboard](#)[All Tickets](#)[New Ticket](#)[API Docs](#)

All Tickets

Showing 1 to 8 of 8 tickets.

Search

Status

Priority

Category

Per page

Search

Reset

ID	Title	Category	Priority	Status	Raised By	Assigned To	Due	Created	Actions
1	VPN drops every few minutes on home Wi-F...	Network	Critical	In Progress	Rahul Sharma	Network Team	Overdue	04 Aug 2026, 11:17	View Edit Delete
8	Install Nodejs LTS and Docker Desktop o...	Software	Low	In Progress	Kavya Reddy	Desktop Support	6 days left	03 Aug 2026, 21:17	View Edit Delete
2	Visual Studio Enterprise license key req...	Software	Medium	Open	Priya Patel	Not assigned	44 hours left	03 Aug 2026, 15:17	View Edit Delete
4	Access request: production analytics das...	Access	High	Open	Sneha Nair	Identity Team	Overdue	02 Aug 2026, 12:17	View Edit Delete
5	Duplicate charge on the SaaS subscriptio...	Billing	High	In Progress	Vikram Desai	Finance Ops	Overdue	01 Aug 2026, 16:17	View Edit Delete
7	Shared drive mapping fails after passwor...	Access	High	Open	Arjun Menon	Not assigned	Overdue	31 Jul 2026, 17:17	View Edit Delete

Figure 4: Search and filter working

Capture: <http://localhost:5185/Ticket/Index?status=Open&priority=High>

localhost:5185/Ticket/Index?status=Open&priority=High

Help Desk Management System

Internal employee support ticket system

[Dashboard](#)[All Tickets](#)[New Ticket](#)[API Docs](#)

All Tickets

Showing 1 to 2 of 2 tickets.

Search

Status

Priority

Category

Per page

Search

Reset

ID	Title	Category	Priority	Status	Raised By	Assigned To	Due	Created	Actions
4	Access request: production analytics das...	Access	High	Open	Sneha Nair	Identity Team	Overdue	02 Aug 2026, 12:17	View Edit Delete
7	Shared drive mapping fails after passwor...	Access	High	Open	Arjun Menon	Not assigned	Overdue	31 Jul 2026, 17:17	View Edit Delete

Help Desk Management System | Rajrup Roy Chowdhury | IN26010404

Figure 5: Ticket details with comment history

Capture: <http://localhost:5185/Ticket/Details/5>

The screenshot shows the 'Ticket 5' details page. The ticket was raised by Vikram Desai on 01 Aug 2026, 16:17. The title is 'Duplicate charge on the SaaS subscription invoice'. The status is 'In Progress' and the priority is 'High'. A red banner indicates 'This ticket is overdue'. The comment history shows a comment from Finance Ops on 01 Aug 2026, 17:17, stating 'Raised a correction with the vendor, reference INV-88421. Expecting the credit note in 2 working days.' The ticket information table shows the status as 'In Progress', priority as 'High', category as 'Billing', raised by 'Vikram Desai', email as 'vikram.desai@example.com', assigned to 'Finance Ops', created on '01 Aug 2026, 16:17', updated on '01 Aug 2026, 16:17', due date as '02 Aug 2026, 16:17', and SLA as 'Overdue'. The actions section includes 'Edit Ticket' and 'Delete Ticket' buttons.

Ticket 5
Raised by Vikram Desai on 01 Aug 2026, 16:17

Duplicate charge on the SaaS subscription invoice

In Progress **High** **This ticket is overdue**

The August invoice lists the design tool subscription twice. Finance needs a corrected invoice before the month-end close.

Comments and History (1)

Finance Ops - 01 Aug 2026, 17:17
Raised a correction with the vendor, reference INV-88421. Expecting the credit note in 2 working days. **Delete**

Add a Comment

Your Name

Comment

Add Comment

Change Status

Open **In Progress** **Resolved** **Closed**

The change will be saved in the history above.

Ticket Information

Status	In Progress
Priority	High
Category	Billing
Raised by	Vikram Desai
Email	vikram.desai@example.com
Assigned to	Finance Ops
Created	01 Aug 2026, 16:17
Updated	01 Aug 2026, 16:17
Due date	02 Aug 2026, 16:17
SLA	Overdue

Actions

Edit Ticket **Delete Ticket**

Figure 6: Raise a new ticket form

Capture: <http://localhost:5185/Ticket/Create>

The screenshot shows the 'Raise a New Ticket' form. The title field has an example: 'Laptop will not connect to the office Wi-Fi'. The description field has a placeholder: 'Explain what the problem is and when it started.' The form includes fields for 'Your Name', 'Your Email', 'Priority' (set to 'Medium'), 'Category' (set to 'Other'), and 'Assign To'. A note at the bottom states: 'New tickets are always created with the status "Open".' The priority dropdown menu shows options: 'Medium', 'High', 'Low', and 'Critical'. The category dropdown menu shows options: 'Other', 'Hardware', 'Software', and 'Network'.

Help Desk Management System
Internal employee support ticket system

Raise a New Ticket
New tickets are always created with the status "Open".

Title *

Example: Laptop will not connect to the office Wi-Fi

Description *

Explain what the problem is and when it started.

Your Name *

Your Email

Priority

Medium

Time to fix: Critical = 4 hours, High = 24 hours, Medium = 72 hours, Low = 7 days.

Category

Other

Assign To

Figure 7: Validation error messages

Capture: Submit the Create form with empty fields

Help Desk Management System
Internal employee support ticket system

Dashboard All Tickets **New Ticket** API Docs

Raise a New Ticket

New tickets are always created with the status "Open".

[Back to list](#)

Title *
Example: Laptop will not connect to the office Wi-Fi

Description *
Explain what the problem is and when it started.

Your Name *

Your Email

Priority
Medium

Time to fix Critical = 4 hours, High = 24 hours, Medium = 72 hours, Low = 7 days.

Category
Other

Assign To

Figure 8: Edit ticket form

Capture: <http://localhost:5185/Ticket/Edit/2>

Help Desk Management System
Internal employee support ticket system

Dashboard All Tickets New Ticket API Docs

Edit Ticket 2

Created on 03 Aug 2026, 15:17

[Back to ticket](#)

Title *
Visual Studio Enterprise license key required

Description *
Starting on the payments migration next week and need an Enterprise license key for profiling and architecture tooling.

Status
Open

Choosing Resolved or Closed will save the date it was completed.

Priority
Medium

Category
Software

Raised By *
Priya Patel

Email

Figure 9: Delete confirmation page

Capture: <http://localhost:5185/Ticket/Delete/3>

The screenshot shows a web browser at localhost:5185/Ticket/Details/9. The page title is "Help Desk Management System" with the subtitle "Internal employee support ticket system". The navigation bar includes "Dashboard", "All Tickets", "New Ticket", and "API Docs". A green message box at the top states "Ticket #9 raised successfully." Below this, the "Ticket 9" section shows it was raised by Rajrup Roy Chowdhury on 04 Aug 2026, 18:33. The ticket description is "Wrong version deployment" with status "Open" and priority "Critical". The description text reads: "A wrong version of our code was updated and deployed, can be a tag mistake needs to be checked until then we need to shutdown the service". To the right, the "Change Status" section has buttons for "Open", "In Progress", "Resolved", and "Closed". Below this is the "Ticket Information" section with details: Status: Open, Priority: Critical, Category: Software, Raised by: Rajrup Roy Chowdhury, Email: rajrup.23bai10213@vitbhopal.ac.in, Assigned to: Sneha, Created: 04 Aug 2026, 18:33, Updated: 04 Aug 2026, 18:33, Due date: 04 Aug 2026, 22:33. At the bottom left, the "Comments and History (1)" section shows a comment from "Help Desk" dated 04 Aug 2026, 18:33: "Ticket raised by Rajrup Roy Chowdhury with Critical priority. SLA due 2026-08-04 17:03 UTC." There is also an "Add a Comment" section with fields for "Your Name" and "Comment". A "Back to list" button is located in the top right of the ticket details area.

Figure 10: Database records (optional)

Capture: *helpdesk.db* opened in DB Browser for SQLite

The screenshot shows the DB Browser for SQLite interface with the 'Comments' table selected. The table has columns: Id, TicketId, Author, Body, IsSystem, and CreatedAt. The data is as follows:

Id	TicketId	Author	Body	IsSystem	CreatedAt
1	1	Network Team	Reproduced on the 6.2 client. Rollin...	0	2026-08-04 06:47:05.7831998
2	2	Hardware Support	Replaced the dock firmware and ..	0	2026-07-30 12:47:05.7831998
3	3	Amit Kumar	No flicker since the swap. Thanks!	0	2026-07-30 13:47:05.7831998
4	4	Finance Ops	Raised a correction with the vendor,...	0	2026-08-01 11:47:05.7831998
5	5	Hardware Support	Battery replaced under warranty and ..	0	2026-07-26 12:47:05.7831998
6	6	Help Desk	Ticket raised by Rajrup Roy Chowdhur...	1	2026-08-04 13:03:17.7518936

The interface also shows a right-hand pane for editing the selected row (row=1, column=0) and a bottom status bar indicating "1 - 6 of 6" records.

4. REST API Endpoint Specifications

Base route: `/api/tickets`

Method	Endpoint	Description
GET	<code>/api/tickets</code>	Retrieve tickets with search, filter, sort and paging
GET	<code>/api/tickets/{id}</code>	Retrieve one ticket with its comment thread
GET	<code>/api/tickets/stats</code>	Retrieve summary counts used by the dashboard
POST	<code>/api/tickets</code>	Create a new ticket (returns 201 Created)
PUT	<code>/api/tickets/{id}</code>	Update an existing ticket
PATCH	<code>/api/tickets/{id}/status</code>	Change only the status of a ticket
DELETE	<code>/api/tickets/{id}</code>	Delete a ticket and its comments (204 No Content)
GET	<code>/api/tickets/{id}/comments</code>	Retrieve all comments for a ticket
POST	<code>/api/tickets/{id}/comments</code>	Add a comment to a ticket
DELETE	<code>/api/tickets/{id}/comments/{commentId}</code>	Delete a user comment
GET	<code>/health</code>	Check the API and database are working

4.1 Query String Parameters

Parameter	Example	Purpose
search	vpn	Matches title, description, requester or assignee
status	Open	Filter by status
priority	Critical	Filter by priority
category	Network	Filter by category
assignedTo	unassigned	Filter by assignee, or list unassigned tickets
overdueOnly	true	Show only tickets past their SLA date
sortBy / sortDir	Priority / Asc	Sorting
page / pageSize	1 / 10	Paging

4.2 Example Request and Response

```
curl "http://localhost:5285/api/tickets?status=Open&priority=High"
```

```
{
  "items": [
    {
      "id": 4,
      "title": "Access request: production analytics dashboard",
      "priority": "High",
      "status": "Open",

```

```
    "category": "Access",
    "raisedBy": "Sneha Nair",
    "assignedTo": "Identity Team",
    "dueDate": "2026-08-03T04:56:17Z",
    "isOverdue": true,
    "commentCount": 0
  }
],
"page": 1,
"pageSize": 10,
"totalCount": 1,
"totalPages": 1
}
```

5. Business Rules Implemented

Rule	Description
Status on creation	Every new ticket is created as Open. The create DTO has no status field, so a client cannot create a ticket that is already closed.
SLA due date	Calculated from priority: Critical = 4 hours, High = 24 hours, Medium = 72 hours, Low = 7 days.
Priority change	If priority changes, the due date is recalculated from the original creation time.
Resolution date	Set when a ticket becomes Resolved or Closed, and cleared again if the ticket is reopened.
History entries	Every status, priority or assignment change adds a system comment recording who made the change.
System comments	Cannot be deleted by users, because they are the audit trail of the ticket.
Deleting a ticket	Also deletes all of its comments (cascade delete).
Time zone handling	All times are stored in UTC and converted to local time only when displayed.

6. Unit Test Execution Results

The test project contains 37 tests covering the service and repository layers, including SLA calculation, status transitions, the history trail, cascade delete, search behaviour, sorting order and paging.

Command used to run the tests:

```
dotnet test
```

Test output:

```
[C:\Users\RAJ\Downloads\ticket mangement system>dotnet run --project src/HelpDesk.Mvc --urls
http://localhost:5185
Building...
info: Microsoft.AspNetCore.DataProtection.KeyManagement.XmlKeyManager[63]
      User profile is available. Using 'C:\Users\RAJ\AppData\Local\ASP.NET\DataProtection-Keys'
as key repository and Windows DPAPI to encrypt keys at rest.
fail: Microsoft.Extensions.Hosting.Internal.Host[11]
      Hosting failed to start
      System.IO.IOException: Failed to bind to address http://127.0.0.1:5185: address already
in use.
      ---> Microsoft.AspNetCore.Connections.AddressInUseException: Only one usage of each
socket address (protocol/network address/port) is normally permitted.
      ---> System.Net.Sockets.SocketException (10048): Only one usage of each socket address
(protocol/network address/port) is normally permitted.
      at System.Net.Sockets.Socket.UpdateStatusAfterSocketErrorAndThrowException(SocketError
error, Boolean disconnectOnFailure, String callerName)
      at System.Net.Sockets.Socket.DoBind(EndPoint endPointSnapshot, SocketAddress
socketAddress)
      at System.Net.Sockets.Socket.Bind(EndPoint localEP)
      at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketTransportOptions.CreateDefaultBound
ListenSocket(EndPoint endpoint)
      at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketConnectionListener.Bind()
      --- End of inner exception stack trace ---
      at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketConnectionListener.Bind()
      at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketTransportFactory.BindAsync(EndPoint
endpoint, CancellationToken cancellationToken)
      at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.Infrastructure.TransportManager.BindAsync(End
Point endPoint, ConnectionDelegate connectionDelegate, EndpointConfig endpointConfig,
CancellationToken cancellationToken)
      at
Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.<>c__DisplayClass28_0`1.<<StartAsync
>g__OnBind|0>d.MoveNext()
      --- End of stack trace from previous location ---
      at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindEndpointAsync(ListenOptions
endpoint, AddressBindContext context, CancellationToken cancellationToken)
      --- End of inner exception stack trace ---
      at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindEndpointAsync(ListenOptions
endpoint, AddressBindContext context, CancellationToken cancellationToken)
      at
Microsoft.AspNetCore.Server.Kestrel.Core.LocalhostListenOptions.BindAsync(AddressBindContext
context, CancellationToken cancellationToken)
      at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.AddressesStrategy.BindAsync(Add
ressBindContext context, CancellationToken cancellationToken)
      at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindAsync(ListenOptions[]
```

```

listenOptions, AddressBindContext context, Func`2 useHttps, CancellationTok
cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.BindAsync(CancellationTok
cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.StartAsync[TContext](IHttpApplicatio
n`1 application, CancellationTok cancellationToken)
    at Microsoft.AspNetCore.Hosting.GenericWebHostService.StartAsync(CancellationTok
cancellationToken)
    at Microsoft.Extensions.Hosting.Internal.Host.<StartAsync>b__15_1(IHostedService
service, CancellationTok token)
    at Microsoft.Extensions.Hosting.Internal.Host.ForeachService[T](IEnumerable`1
services, CancellationTok token, Boolean concurrent, Boolean abortOnFirstException, List`1
exceptions, Func`3 operation)
Unhandled exception. System.IO.IOException: Failed to bind to address http://127.0.0.1:5185:
address already in use.
---> Microsoft.AspNetCore.Connections.AddressInUseException: Only one usage of each socket
address (protocol/network address/port) is normally permitted.
---> System.Net.Sockets.SocketException (10048): Only one usage of each socket address
(protocol/network address/port) is normally permitted.
    at System.Net.Sockets.Socket.UpdateStatusAfterSocketErrorAndThrowException(SocketError
error, Boolean disconnectOnFailure, String callerName)
    at System.Net.Sockets.Socket.DoBind(EndPoint endPointSnapshot, SocketAddress socketAddress)
    at System.Net.Sockets.Socket.Bind(EndPoint localEP)
    at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketTransportOptions.CreateDefaultBound
ListenSocket(EndPoint endPoint)
    at Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketConnectionListener.Bind()
    --- End of inner exception stack trace ---
    at Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketConnectionListener.Bind()
    at
Microsoft.AspNetCore.Server.Kestrel.Transport.Sockets.SocketTransportFactory.BindAsync(EndPoint
endpoint, CancellationTok cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.Infrastructure.TransportManager.BindAsync(End
Point endPoint, ConnectionDelegate connectionDelegate, EndpointConfig endpointConfig,
CancellationTok cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.<>c__DisplayClass28_0`1.<<StartAsync
>g__OnBind|0>d.MoveNext()
    --- End of stack trace from previous location ---
    at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindEndpointAsync(ListenOptions
endpoint, AddressBindContext context, CancellationTok cancellationToken)
    --- End of inner exception stack trace ---
    at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindEndpointAsync(ListenOptions
endpoint, AddressBindContext context, CancellationTok cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.LocalhostListenOptions.BindAsync(AddressBindContext
context, CancellationTok cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.AddressesStrategy.BindAsync(Add
ressBindContext context, CancellationTok cancellationToken)
    at Microsoft.AspNetCore.Server.Kestrel.Core.Internal.AddressBinder.BindAsync(ListenOptions[]
listenOptions, AddressBindContext context, Func`2 useHttps, CancellationTok
cancellationToken)
    at Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.BindAsync(CancellationTok
cancellationToken)
    at
Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServerImpl.StartAsync[TContext](IHttpApplicatio
n`1 application, CancellationTok cancellationToken)
    at Microsoft.AspNetCore.Hosting.GenericWebHostService.StartAsync(CancellationTok
cancellationToken)
    at Microsoft.Extensions.Hosting.Internal.Host.<StartAsync>b__15_1(IHostedService service,
CancellationTok token)

```

```
at Microsoft.Extensions.Hosting.Internal.Host.ForeachService[T](IEnumerable`1 services,
Cancellation token, Boolean concurrent, Boolean abortOnFirstException, List`1 exceptions,
Func`3 operation)
at Microsoft.Extensions.Hosting.Internal.Host.StartAsync(CancellationToken
cancellationToken)
at Microsoft.Extensions.Hosting.HostingAbstractionsHostExtensions.RunAsync(IHost host,
Cancellation token)
at Microsoft.Extensions.Hosting.HostingAbstractionsHostExtensions.RunAsync(IHost host,
Cancellation token)
at Microsoft.Extensions.Hosting.HostingAbstractionsHostExtensions.Run(IHost host)
at Program.<Main>$(String[] args) in C:\Users\RAJ\Downloads\ticket mangement
system\src\HelpDesk.Mvc\Program.cs:line 38

C:\Users\RAJ\Downloads\ticket mangement system>dotnet test
Determining projects to restore...
All projects are up-to-date for restore.
HelpDesk.Contracts -> C:\Users\RAJ\Downloads\ticket mangement
system\src\HelpDesk.Contracts\bin\Debug\net8.0\HelpDesk
.Contracts.dll
HelpDesk.Api -> C:\Users\RAJ\Downloads\ticket mangement
system\src\HelpDesk.Api\bin\Debug\net8.0\HelpDesk.Api.dll
HelpDesk.Tests -> C:\Users\RAJ\Downloads\ticket mangement
system\tests\HelpDesk.Tests\bin\Debug\net8.0\HelpDesk.Tests
.dll
Test run for C:\Users\RAJ\Downloads\ticket mangement
system\tests\HelpDesk.Tests\bin\Debug\net8.0\HelpDesk.Tests.dll (.NETCoreApp,Version=v8.0)
VSTest version 17.11.1 (x64)

Starting test execution, please wait...
A total of 1 test files matched the specified pattern.

Passed! - Failed:    0, Passed:    37, Skipped:    0, Total:    37, Duration: 471 ms -
HelpDesk.Tests.dll (net8.0)
```

Figure 11: Unit test results

Capture: Terminal output of dotnet test

[Paste Figure 11 screenshot here]

6.1 Test Summary

Test Area	Tests	Result
SLA due date calculation	5	
Status transitions and resolution date	4	
History / audit trail	3	
Create and update rules	5	
Comments	4	
Statistics	4	

Test Area	Tests	Result
Search, filter, sort and paging	12	
Total	37	

7. How to Run the Project

Requirement: .NET 8 SDK. No database installation is needed — the SQLite file is created and filled with sample data automatically the first time the API runs.

Step 1 — start the Web API:

```
dotnet run --project src/HelpDesk.Api --urls http://localhost:5285
```

Step 2 — start the MVC website (second terminal):

```
dotnet run --project src/HelpDesk.Mvc --urls http://localhost:5185
```

Step 3 — open in the browser:

Application	Address
Help Desk website	http://localhost:5185
API documentation	http://localhost:5285/swagger
API health check	http://localhost:5285/health

The Web API must be started first, because the website reads all of its data from the API.

8. Source Code Listings

The complete source code is available in the repository. The main files are listed below.

```
HelpDesk.sln
+-- src/
|   +-- HelpDesk.Contracts/
|   |   +-- Enums.cs                (TicketStatus, TicketPriority, TicketCategory)
|   |   +-- TicketDtos.cs          (Create / Update / Response DTOs)
|   |   +-- QueryDtos.cs           (TicketQuery, PagedResult, TicketStatsDto)
|   +-- HelpDesk.Api/
|   |   +-- Models/Ticket.cs
|   |   +-- Models/TicketComment.cs
|   |   +-- Models/SlaPolicy.cs
|   |   +-- Data/HelpDeskDbContext.cs
|   |   +-- Data/DbSeeder.cs
|   |   +-- Repositories/ITicketRepository.cs
|   |   +-- Repositories/TicketRepository.cs
|   |   +-- Services/ITicketService.cs
|   |   +-- Services/TicketService.cs
|   |   +-- Controllers/TicketsController.cs
|   |   +-- Program.cs
|   +-- HelpDesk.Mvc/
|   |   +-- Controllers/TicketController.cs
|   |   +-- Services/TicketApiClient.cs
|   |   +-- Models/Display.cs
|   |   +-- Views/Ticket/*.cshtml
|   |   +-- wwwroot/css/site.css
+-- tests/
    +-- HelpDesk.Tests/
        +-- TicketServiceTests.cs
        +-- TicketRepositoryTests.cs
```

8.1 HelpDesk.Contracts / Enums.cs

```
namespace HelpDesk.Contracts;

public enum TicketStatus
{
    Open,
    InProgress,
    Resolved,
    Closed
}

public enum TicketPriority
{
    Low,
    Medium,
    High,
    Critical
}

public enum TicketCategory
{
    Hardware,
    Software,
    Network,
    Access,
    Billing,
    Other
}
```

8.2 HelpDesk.Api / Models / SlaPolicy.cs

```
namespace HelpDesk.Api.Models;

public static class SlaPolicy
{
    public static TimeSpan ResolutionWindowFor(TicketPriority priority) => priority
    switch
    {
        {
            TicketPriority.Critical => TimeSpan.FromHours(4),
            TicketPriority.High     => TimeSpan.FromHours(24),
            TicketPriority.Medium   => TimeSpan.FromHours(72),
            TicketPriority.Low      => TimeSpan.FromDays(7),
            -                       => TimeSpan.FromDays(7)
        };

    public static DateTime DueDateFor(TicketPriority priority, DateTime createdAtUtc)
    =>
        createdAtUtc.Add(ResolutionWindowFor(priority));
}
```

8.3 HelpDesk.Api / Repositories / ITicketRepository.cs

```
public interface ITicketRepository
{
    Task<(IReadOnlyList<Ticket> Items, int TotalCount)> QueryAsync(
        TicketQuery query, CancellationToken ct = default);

    Task<Ticket?> GetByIdAsync(int id, bool includeComments = false,
        bool tracked = false, CancellationToken ct = default);

    Task<Ticket> AddAsync(Ticket ticket, CancellationToken ct = default);
    Task SaveChangesAsync(CancellationToken ct = default);
    Task<bool> DeleteAsync(int id, CancellationToken ct = default);
    Task<bool> ExistsAsync(int id, CancellationToken ct = default);
    Task<IReadOnlyList<Ticket>> GetAllAsync(CancellationToken ct = default);

    Task<TicketComment> AddCommentAsync(TicketComment comment, CancellationToken ct =
    default);
    Task<IReadOnlyList<TicketComment>> GetCommentsAsync(int ticketId, CancellationToken
    ct = default);
    Task<TicketComment?> GetCommentAsync(int ticketId, int commentId, CancellationToken
    ct = default);
    Task<bool> DeleteCommentAsync(int ticketId, int commentId, CancellationToken ct =
    default);
}
```

9. Conclusion

The Help Desk Management System meets all of the requirements set for the assignment. A REST Web API was built using ASP.NET Core and Entity Framework Core with the Repository Pattern, and a separate MVC web application consumes that API to provide the user interface.

Beyond the basic CRUD requirement, the project also implements searching, filtering, sorting, paging, SLA tracking, a comment and history system, and a dashboard with summary reporting. Validation is enforced on both the client and the server using a shared contracts project, so the rules cannot get out of step between the two applications. Unit tests were written for the business rules to confirm that the SLA calculations and status transitions behave correctly.

Possible future improvements:

- User login and roles, so that employees and agents see different screens
- Email notifications when a ticket is assigned or resolved
- File attachments on tickets
- Moving from EnsureCreated() to Entity Framework migrations for schema changes